

Martingales — Solutions (set 2)

Problem I — Records of an i.i.d. sequence: a martingale and a weak law of large numbers (24 pts)

(a) (M_n) is a martingale.

Integrability and adaptedness. For each n , $N_n = \sum_{k=1}^n R_k$ is a finite sum of indicators, so $0 \leq N_n \leq n$ and $M_n = N_n - H_n$ is bounded, in particular integrable. Moreover N_n is $\sigma(R_1, \dots, R_n) = \mathcal{F}_n$ -measurable, and H_n is deterministic, so M_n is $\mathcal{F}_n$ -measurable. Thus (M_n) is adapted and integrable.

Martingale property. For $n \geq 1$,

$$M_{n+1} - M_n = (N_{n+1} - N_n) - (H_{n+1} - H_n) = R_{n+1} - \frac{1}{n+1}.$$

Now R_{n+1} is independent of $(R_1, \dots, R_n)$, hence of $\mathcal{F}_n$, and $\mathbb{E}[R_{n+1}] = \frac{1}{n+1}$. Consequently

$$\mathbb{E}[M_{n+1} - M_n \mid \mathcal{F}_n] = \mathbb{E}\left[R_{n+1} - \frac{1}{n+1} \mid \mathcal{F}_n\right] = \mathbb{E}[R_{n+1}] - \frac{1}{n+1} = 0,$$

where we used that $R_{n+1} \perp \mathcal{F}_n$ implies $\mathbb{E}[R_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[R_{n+1}]$. Hence $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$, and $(M_n)_{n \geq 1}$ is an $(\mathcal{F}_n)$ -martingale. $\square$

(b) **Computation of $\mathbb{E}[N_n \mid \mathcal{F}_m]$ for $m \leq n$.**

Write $N_n = N_m + \sum_{k=m+1}^n R_k$. The variable N_m is $\mathcal{F}_m$ -measurable, so

$$\mathbb{E}[N_n \mid \mathcal{F}_m] = N_m + \sum_{k=m+1}^n \mathbb{E}[R_k \mid \mathcal{F}_m].$$

For $k > m$, R_k is independent of $\mathcal{F}_m = \sigma(R_1, \dots, R_m)$ (independence of the R_j), whence $\mathbb{E}[R_k \mid \mathcal{F}_m] = \mathbb{E}[R_k] = \frac{1}{k}$. Therefore

$$\mathbb{E}[N_n \mid \mathcal{F}_m] = N_m + \sum_{k=m+1}^n \frac{1}{k} = N_m + (H_n - H_m).$$

(Equivalently, $\mathbb{E}[M_n \mid \mathcal{F}_m] = M_m$ gives back $\mathbb{E}[N_n \mid \mathcal{F}_m] - H_n = N_m - H_m$, consistent with (a).) $\square$

(c) **Variance of N_n and law of large numbers.**

Since the R_k are independent and $R_k \sim \text{Bernoulli}(1/k)$,

$$\text{Var}(R_k) = \frac{1}{k} \left(1 - \frac{1}{k}\right) = \frac{1}{k} - \frac{1}{k^2}.$$

By independence, the variances add up:

$$\text{Var}(N_n) = \sum_{k=1}^n \text{Var}(R_k) = \sum_{k=1}^n \frac{1}{k} - \sum_{k=1}^n \frac{1}{k^2} = H_n - \sum_{k=1}^n \frac{1}{k^2}.$$

In particular $0 \leq \text{Var}(N_n) \leq H_n$, and $\text{Var}(N_n) = H_n + O(1) = \log n + O(1)$, since $\sum_{k \geq 1} k^{-2} = \pi^2/6 < \infty$.

Expectation. $\mathbb{E}[N_n] = \sum_{k=1}^n \frac{1}{k} = H_n = \log n + O(1)$.

Convergence in probability. Fix $\varepsilon > 0$. By the Bienaymé–Chebyshev inequality applied to N_n , centered at its mean H_n ,

$$\mathbb{P}\left(\left|\frac{N_n}{H_n} - 1\right| > \varepsilon\right) = \mathbb{P}(|N_n - H_n| > \varepsilon H_n) \leq \frac{\text{Var}(N_n)}{\varepsilon^2 H_n^2} \leq \frac{H_n}{\varepsilon^2 H_n^2} = \frac{1}{\varepsilon^2 H_n} \xrightarrow{n \rightarrow \infty} 0,$$

since $H_n \rightarrow \infty$. Hence $\frac{N_n}{H_n} \xrightarrow{\mathbb{P}} 1$.

Finally, $\frac{H_n}{\log n} \rightarrow 1$ (because $H_n = \log n + O(1)$), so by taking the product

$$\frac{N_n}{\log n} = \frac{N_n}{H_n} \cdot \frac{H_n}{\log n} \xrightarrow{\mathbb{P}} 1 \cdot 1 = 1.$$

(The product of a sequence converging in probability to 1 with a deterministic sequence tending to 1 converges in probability to 1.) $\square$

Problem II — The Gambling-Team Method and the Expected Waiting Time for the Pattern HTH (26 pts)

(a) (G_n) is a martingale and T a stopping time.

For the player who entered at toss k , we track his fortune. As long as he bets, at each toss he stakes his entire fortune on an outcome of probability $\frac{1}{2}$ which doubles in case of success and vanishes otherwise; this is a fair game. Let $W_n^{(k)}$ denote the fortune (final or current) of player k after toss n (with $W_n^{(k)} = 1$ for $n < k$, his initial stake being accounted for separately). Conditionally on $\mathcal{F}_{n-1}$, if he is still betting at toss n then $W_n^{(k)} = 2W_{n-1}^{(k)}$ with probability $\frac{1}{2}$ and $W_n^{(k)} = 0$ with probability $\frac{1}{2}$, hence $\mathbb{E}[W_n^{(k)} \mid \mathcal{F}_{n-1}] = W_{n-1}^{(k)}$; if he has already finished betting (ruined or having completed the pattern), $W_n^{(k)} = W_{n-1}^{(k)}$. In all cases

$$\mathbb{E}[W_n^{(k)} \mid \mathcal{F}_{n-1}] = W_{n-1}^{(k)}.$$

The cumulative net gain is written $G_n = \sum_{k=1}^n (W_n^{(k)} - 1)$ (each player who entered contributes a stake 1 and holds a fortune $W_n^{(k)}$). It is a finite sum of $\mathcal{F}_n$ -measurable variables, and each individual fortune satisfies $0 \leq W_n^{(k)} \leq 8$ (at most three consecutive successes 2^3), whence $|G_n| \leq \sum_{k=1}^n W_n^{(k)} \leq 8n$, so $G_n \in L^1$. Finally, since exactly one new player (with stake 1, initial fortune $W_{n-1}^{(n)} = 1$) enters at toss n ,

$$\mathbb{E}[G_n \mid \mathcal{F}_{n-1}] = \sum_{k=1}^n (\mathbb{E}[W_n^{(k)} \mid \mathcal{F}_{n-1}] - 1) = \sum_{k=1}^n (W_{n-1}^{(k)} - 1) = G_{n-1},$$

where we used $W_{n-1}^{(n)} - 1 = 0$. Therefore $(G_n)_{n \geq 0}$ is an $(\mathcal{F}_n)$ -martingale, with $\mathbb{E}[G_n] = G_0 = 0$.

For T : the event $\{T = n\}$ is written $\{(X_{n-2}, X_{n-1}, X_n) = (\text{P}, \text{F}, \text{P})\} \cap \bigcap_{3 \leq m < n} \{(X_{m-2}, X_{m-1}, X_m) \neq (\text{P}, \text{F}, \text{P})\}$ and hence depends only on $(X_1, \dots, X_n)$, whence $\{T = n\} \in \mathcal{F}_n$ for all n . Thus T is a stopping time.

(b) **Value of G_T .**

At time T , we have $(X_{T-2}, X_{T-1}, X_T) = (\text{P}, \text{F}, \text{P})$. Let us examine the players still present (nonzero fortune) among the most recent ones:

- *Player who entered at toss $T - 2$:* he bet P at toss $T - 2$ (success since $X_{T-2} = \text{P}$), F at toss $T - 1$ (success since $X_{T-1} = \text{F}$), P at toss T (success since $X_T = \text{P}$). He has completed the pattern PFP: his final fortune is $2^3 = 8$.
- *Player who entered at toss $T - 1$:* he bet P at toss $T - 1$, but $X_{T-1} = \text{F}$: he is ruined, fortune 0.
- *Player who entered at toss T :* he bet P at toss T (success since $X_T = \text{P}$). He is still in the game (he has just succeeded in the first bet), his current fortune at time T is 2. (This is where the self-overlap comes into play: the suffix P of the pattern is also its prefix, so this player has “started” the pattern.)
- All the players who entered at a toss $\leq T - 3$ have, by definition of T , been ruined before time T : if one of them were still betting, the corresponding tosses would have had to reproduce a coinciding beginning of the pattern, and the only possibility of a complete pattern earlier is excluded by the minimality of T ; in any case none has completed the pattern before T , and a direct computation shows that their fortune is 0 at time T . (Concretely, since $X_{T-2} = \text{P}$, only a player who entered at $T - 2, T - 1, T$ can be active: a player who entered earlier would have been ruined at the latest at toss $T - 2$ if he was betting on P, or before.)

The total final fortune at time T is therefore $8 + 0 + 2 = 10$. The number of players who entered up to time T is T (one per toss), so the total of the stakes is T . Hence

$$\boxed{G_T = 10 - T.}$$

(c) Computation of $\mathbb{E}[T]$.

We apply the optional stopping theorem to the martingale (G_n) and the stopping time T . The hypotheses are satisfied:

- $\mathbb{E}[T] < \infty$ (given), so $T < \infty$ a.s.;
- the increments of G are bounded: at toss n , the variation $G_n - G_{n-1} = \sum_{k \leq n} (W_n^{(k)} - W_{n-1}^{(k)})$ involves only the (at most) three players still active (who entered at $n-2, n-1, n$), each seeing his fortune go from a value ≤ 4 to a value ≤ 8 or to 0, hence vary by at most 8 in absolute value; thus $|G_n - G_{n-1}| \leq C$ with $C = 24$ (constant).

Under these hypotheses (bounded increments and $\mathbb{E}[T] < \infty$), the optional stopping theorem gives

$$\mathbb{E}[G_T] = \mathbb{E}[G_0] = 0.$$

By part (b), $G_T = 10 - T$, whence $\mathbb{E}[10 - T] = 0$, that is,

$$\boxed{\mathbb{E}[T] = 10.}$$

Remark (verification). The value $10 = 2^3 + 2^1$ corresponds to “Conway’s formula”: one adds 2^ℓ for each length ℓ such that a suffix of the pattern of length ℓ coincides with its prefix. For PFP, the overlaps are of lengths 3 (the whole pattern) and 1 (the final P = the initial P), whence $2^3 + 2^1 = 10$. For a pattern of length 3 without self-overlap such as PPF, one would have only $\mathbb{E}[T] = 2^3 = 8$, which illustrates the effect of the structure of the pattern on the mean waiting time.

Problem III — Random series with random signs: an L^2 -bounded martingale (25 pts)

(a) Martingale bounded in L^2 .

Integrability and measurability. For each n , X_n is a finite sum of bounded random variables, hence X_n is bounded, in particular $X_n \in L^1$ (and even $X_n \in L^2$), and X_n is $\mathcal{F}_n$ -measurable as a function of $\varepsilon_1, \dots, \varepsilon_n$.

Martingale property. For $n \geq 0$, since ε_{n+1} is independent of $\mathcal{F}_n$ and has zero mean,

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[X_n + a_{n+1}\varepsilon_{n+1} \mid \mathcal{F}_n] = X_n + a_{n+1} \mathbb{E}[\varepsilon_{n+1} \mid \mathcal{F}_n] = X_n + a_{n+1} \mathbb{E}[\varepsilon_{n+1}] = X_n$$

almost surely, where we used that X_n is $\mathcal{F}_n$ -measurable and that $\mathbb{E}[\varepsilon_{n+1}] = 0$. Hence $(X_n)_{n \geq 0}$ is a martingale.

L^2 bound. The variables $a_k \varepsilon_k$ are centered and independent (hence pairwise uncorrelated); moreover $\mathbb{E}[(a_k \varepsilon_k)^2] = a_k^2 \mathbb{E}[\varepsilon_k^2] = a_k^2$. By orthogonality in L^2 ,

$$\mathbb{E}[X_n^2] = \mathbb{E}\left[\left(\sum_{k=1}^n a_k \varepsilon_k\right)^2\right] = \sum_{k=1}^n a_k^2 + \sum_{1 \leq j \neq k \leq n} a_j a_k \mathbb{E}[\varepsilon_j \varepsilon_k] = \sum_{k=1}^n a_k^2,$$

since for $j \neq k$, $\mathbb{E}[\varepsilon_j \varepsilon_k] = \mathbb{E}[\varepsilon_j] \mathbb{E}[\varepsilon_k] = 0$ by independence. The sequence $(\sum_{k=1}^n a_k^2)_n$ is nondecreasing with limit $\sum_{k \geq 1} a_k^2 < \infty$, hence

$$\sup_{n \geq 0} \mathbb{E}[X_n^2] = \sum_{k \geq 1} a_k^2 < \infty.$$

Thus (X_n) is bounded in L^2 .

(b) Almost sure and L^2 convergence.

By (a), (X_n) is a martingale bounded in L^2 . By the convergence theorem for martingales bounded in L^2 , there exists a random variable $X_\infty \in L^2$ such that

$$X_n \xrightarrow[n \rightarrow \infty]{} X_\infty \quad \text{almost surely and in } L^2.$$

Expectation. The L^2 convergence implies L^1 convergence, hence $\mathbb{E}[X_n] \rightarrow \mathbb{E}[X_\infty]$. But $\mathbb{E}[X_n] = \mathbb{E}[X_0] = 0$ for all n (martingale property), whence

$$\mathbb{E}[X_\infty] = 0.$$

Second moment. Convergence in L^2 implies convergence of the L^2 norms: $\|X_n\|_{L^2} \rightarrow \|X_\infty\|_{L^2}$, hence $\mathbb{E}[X_n^2] \rightarrow \mathbb{E}[X_\infty^2]$. Together with (a),

$$\mathbb{E}[X_\infty^2] = \lim_{n \rightarrow \infty} \mathbb{E}[X_n^2] = \sum_{k \geq 1} a_k^2.$$

(c) Maximal inequality.

Fix $N \geq 1$. The martingale $(X_n)_{0 \leq n \leq N}$ is square integrable. Since the function $t \mapsto t^2$ is nondecreasing on $\mathbb{R}_+$, we have $\max_{0 \leq n \leq N} X_n^2 = (\max_{0 \leq n \leq N} |X_n|)^2 = (S_N^*)^2$. Doob's L^2 maximal inequality therefore gives

$$\mathbb{E}[(S_N^*)^2] = \mathbb{E}\left[\max_{0 \leq n \leq N} X_n^2\right] \leq 4 \mathbb{E}[X_N^2] = 4 \sum_{k=1}^N a_k^2 \leq 4 \sum_{k \geq 1} a_k^2,$$

using (a) for $\mathbb{E}[X_N^2]$ and the nonnegativity of the a_k^2 .

The sequence $((S_N^*)^2)_{N \geq 1}$ is nondecreasing (the maximum is taken over an increasing set of indices) and converges pointwise to $(\sup_{n \geq 0} |X_n|)^2$. By the monotone convergence theorem,

$$\mathbb{E}\left[\left(\sup_{n \geq 0} |X_n|\right)^2\right] = \lim_{N \rightarrow \infty} \mathbb{E}[(S_N^*)^2] \leq 4 \sum_{k \geq 1} a_k^2 < \infty.$$

Since $\sum_{k \geq 1} a_k^2 < \infty$, the random variable $(\sup_{n \geq 0} |X_n|)^2$ has finite expectation, hence is finite almost surely; consequently

$$\sup_{n \geq 0} |X_n| < \infty \quad \text{almost surely.} \quad \square$$

Problem IV — Exit Times and the Reflection Principle for Brownian Motion (25 pts)

1. Exit probabilities by symmetry.

Set $\tilde{B}_t = -B_t$. By hypothesis $(\tilde{B}_t)_{t \geq 0}$ is a standard Brownian motion for the same filtration. Since $|\tilde{B}_t| = |B_t|$ for all t , the exit time associated with $\tilde{B}$,

$$\tilde{\tau} = \inf\{t \geq 0 : |\tilde{B}_t| \geq a\} = \inf\{t \geq 0 : |B_t| \geq a\} = \tau,$$

coincides with τ . Moreover, on the probability-1 event where $\tau < \infty$ (F1) and $B_\tau \in \{-a, a\}$ (F2), we have $\tilde{B}_\tau = -B_\tau$; hence

$$\{B_\tau = a\} = \{\tilde{B}_\tau = -a\} \quad \text{and} \quad \{B_\tau = -a\} = \{\tilde{B}_\tau = a\}.$$

Now $\tilde{B}$ being a standard Brownian motion *with the same law* as B , the pair $(\tilde{\tau}, \tilde{B}_\tau)$ has the same law as (τ, B_τ) . In particular

$$\mathbb{P}(\tilde{B}_\tau = -a) = \mathbb{P}(B_\tau = -a).$$

Combining the last two lines,

$$\mathbb{P}(B_\tau = a) = \mathbb{P}(\tilde{B}_\tau = -a) = \mathbb{P}(B_\tau = -a).$$

By (F1)–(F2) the events $\{B_\tau = a\}$ and $\{B_\tau = -a\}$ partition Ω up to a negligible set, so their probabilities sum to 1. Being equal, each equals $\frac{1}{2}$:

$$\boxed{\mathbb{P}(B_\tau = a) = \mathbb{P}(B_\tau = -a) = \frac{1}{2}.}$$

2. Expectation of the exit time.

The time τ is a stopping time, so $\tau \wedge t$ is a *bounded* stopping time for every $t \geq 0$. The optional stopping theorem applied to the martingale $(B_s^2 - s)_{s \geq 0}$ at the time $\tau \wedge t$ gives

$$\mathbb{E}[B_{\tau \wedge t}^2 - (\tau \wedge t)] = \mathbb{E}[B_0^2 - 0] = 0,$$

that is,

$$\mathbb{E}[\tau \wedge t] = \mathbb{E}[B_{\tau \wedge t}^2] \quad \text{for all } t \geq 0. \quad (*)$$

Left-hand side. The map $t \mapsto \tau \wedge t$ is nondecreasing and tends to τ as $t \rightarrow \infty$; the variables $\tau \wedge t \geq 0$ increase pointwise to τ . By the monotone convergence theorem,

$$\mathbb{E}[\tau \wedge t] \xrightarrow[t \rightarrow \infty]{} \mathbb{E}[\tau],$$

a finite quantity by (F1).

Right-hand side. For $t < \tau$ we have $|B_{\tau \wedge t}| = |B_t| \leq a$ (definition of τ), and for $t \geq \tau$ we have $|B_{\tau \wedge t}| = |B_\tau| = a$; thus $0 \leq B_{\tau \wedge t}^2 \leq a^2$ for all t . Since $\tau < \infty$ a.s. and the trajectories are continuous, $B_{\tau \wedge t} \rightarrow B_\tau$ a.s., hence $B_{\tau \wedge t}^2 \rightarrow B_\tau^2$ a.s. By dominated convergence (domination by the integrable constant a^2),

$$\mathbb{E}[B_{\tau \wedge t}^2] \xrightarrow[t \rightarrow \infty]{} \mathbb{E}[B_\tau^2].$$

By question 1, $B_\tau^2 = a^2$ almost surely (whether $B_\tau = a$ or $B_\tau = -a$), so $\mathbb{E}[B_\tau^2] = a^2$.

Passing to the limit $t \rightarrow \infty$ in (*), we obtain

$$\boxed{\mathbb{E}[\tau] = a^2.}$$

3. Reflection principle and law of M_t .

Explicit form of the tail. We start from the given identity: for $b > 0$,

$$\mathbb{P}\left(\max_{s \in [0, t]} B_s \geq b\right) = 2\mathbb{P}(B_t \geq b).$$

Since $B_t \sim \mathcal{N}(0, t)$, by scaling invariance $B_t/\sqrt{t} \sim \mathcal{N}(0, 1)$, whence

$$\mathbb{P}(B_t \geq b) = \mathbb{P}\left(\frac{B_t}{\sqrt{t}} \geq \frac{b}{\sqrt{t}}\right) = 1 - \Phi(b/\sqrt{t}).$$

(Equivalently, by scaling invariance the process $(t^{-1/2}B_{tu})_{u \geq 0}$ is a standard Brownian motion, which reduces the computation to $u = 1$.) We deduce

$$\boxed{\mathbb{P}\left(\max_{s \in [0, t]} B_s \geq b\right) = 2(1 - \Phi(b/\sqrt{t})), \quad b > 0.}$$

Law of M_t and link with $|B_t|$. With $M_t = \max_{s \in [0, t]} B_s$, the previous equality gives for $b > 0$

$$\mathbb{P}(M_t \geq b) = 2(1 - \Phi(b/\sqrt{t})).$$

On the other hand, by symmetry of the $\mathcal{N}(0, t)$ law, for $b > 0$,

$$\mathbb{P}(|B_t| \geq b) = \mathbb{P}(B_t \geq b) + \mathbb{P}(B_t \leq -b) = 2\mathbb{P}(B_t \geq b) = 2(1 - \Phi(b/\sqrt{t})).$$

The two survival functions therefore coincide on $]0, \infty[$; since $M_t \geq B_0 = 0$ and $|B_t| \geq 0$ almost surely, the two laws are equal:

$$\boxed{M_t \stackrel{(d)}{=} |B_t|.}$$

In other words, M_t follows the “half-normal” law of scale $\sqrt{t}$: its cumulative distribution function is, for $x \geq 0$, $\mathbb{P}(M_t \leq x) = 2\Phi(x/\sqrt{t}) - 1$, and its density is obtained by differentiation,

$$f_{M_t}(x) = \frac{2}{\sqrt{t}} \varphi(x/\sqrt{t}) \mathbf{1}_{\{x \geq 0\}} = \sqrt{\frac{2}{\pi t}} e^{-x^2/(2t)} \mathbf{1}_{\{x \geq 0\}},$$

where $\varphi(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$. $\square$